

xMCF v3.1.1

ISO/PAS 8329

Description of **mechanical** connections and joints in structural systems

ISO/PAS 8329 xMCF at prostep ivip SSB
2025-10-16

This type of connections ...

... not that type!

Agenda

- 1 Joining Technologies & Processes are challenging ...
- 2 χ MCF – the Enabler for Smooth Processes
- 3 χ MCF Use Cases
- 4 History, Status, Outlook & Supporting Organizations
- 5 Status of ISO Standardization
- 6 Strategic Position of χ MCF as a Standard
- 7 Optional: Federated use of STEP AP242 with χ MCF *and* Technical Details of χ MCF



Joining Technologies & Processes are challenging ...

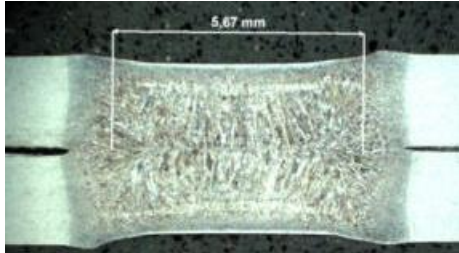
Source of image: <https://fabianoefner.com/disintegrating-2/>



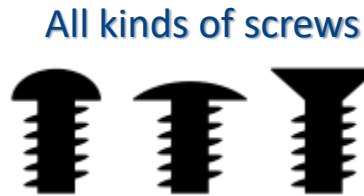
Challenges wrt. Connection Information

Big Variety & Complexity

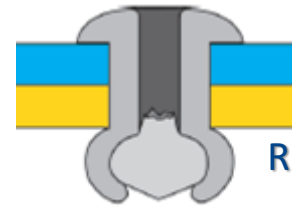
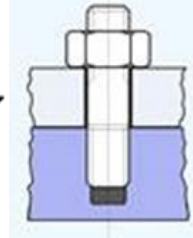
- > 60 known connection techniques.
- Up to 25 quality criteria per connection technique.



Spot Weld (Section)



All kinds of screws



Rivets



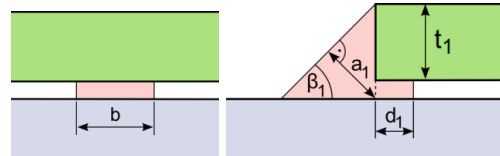
Nails



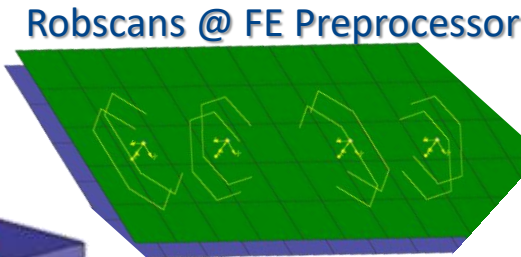
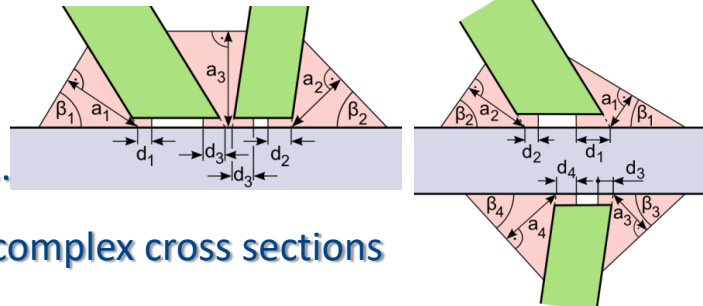
Clips



Clinch



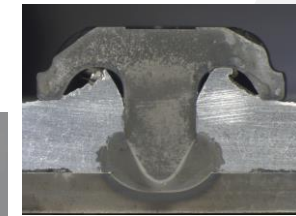
Heat Stake



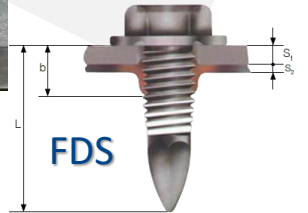
Robscans @ FE Preprocessor



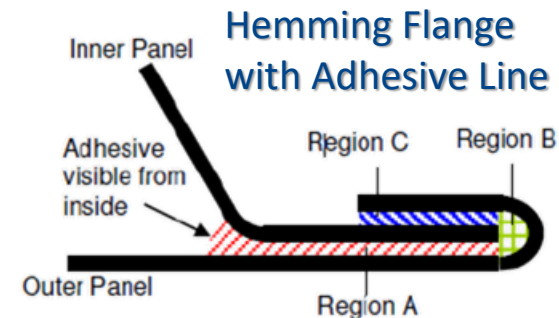
Clinch Rivet Stud



ROTAV



FDS



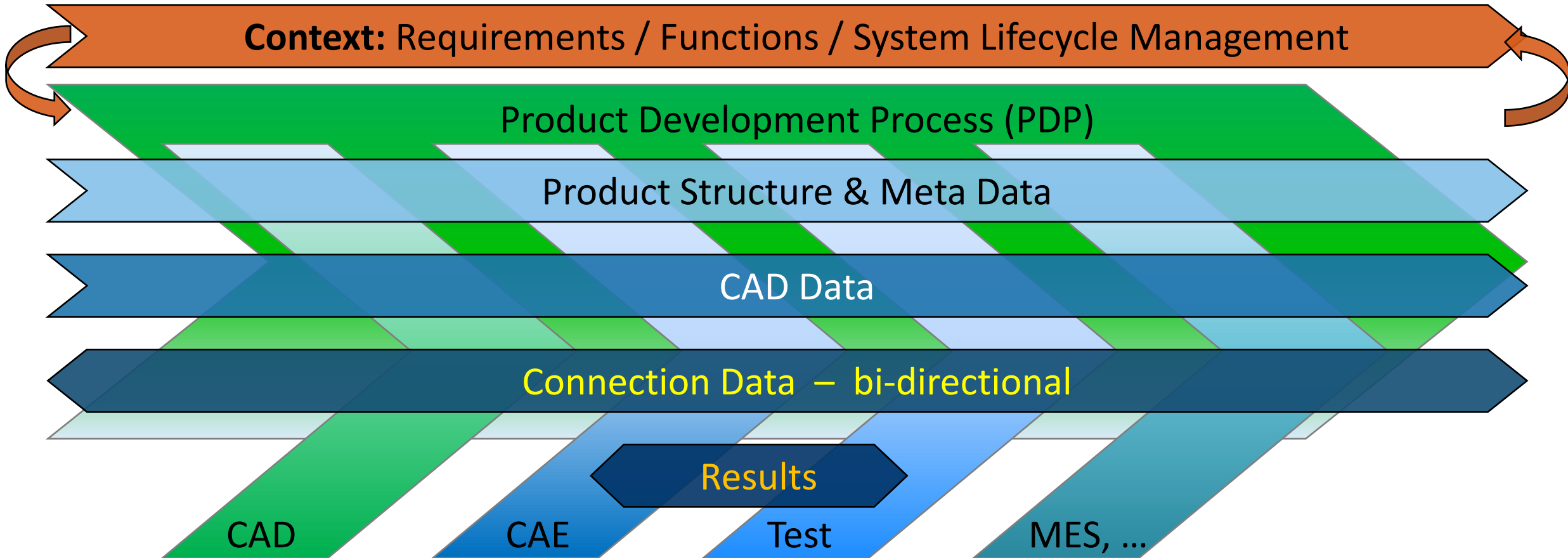
Hemming Flange with Adhesive Line



Adhesive Face

Weld lines with increasingly complex cross sections

Challenges wrt. Product Development Process



Connections are the mandatory **3rd cross-sectional data track** of PDP!

What is so Special about Connection Data?

Connections differ from product structure, meta data and CAD data, since e. g.:

function dominates
over their shape

need more
PLM upstream
data propagation
than CAD

work needs different
tools & plugins,
special process steps
and expert *knowledge*

belong to
inner nodes
of product tree,
not to its leaves

data size is
much smaller
than CAD data

CAD and connections
complement each other.
Each is *useless without*
the other.

What are the Frequent Problems?

- Every OEM creates *own CATIA/NX macros* or buys *proprietary software*.
 - Common suppliers *need to be familiar with all those tools*.
 - Data exchange along process chain needs *additional tools*, frequently “home-brewed”.
⇒ Expensive and error-prone.
 - However, in reality only *few techniques* are supported with only a *fraction of their data*.
 - Inventing new techniques or adding new parameters results in *excessive costs* and *process threats*.
 - Changing software vendors implies *high investments*.
- ⇒ Resulting “*vendor lock-in-effect*” impedes competition and hence *hinders progress*.

χ MCF – the Enabler for Smooth Processes



χMCF: *The* Standard for Connection Information

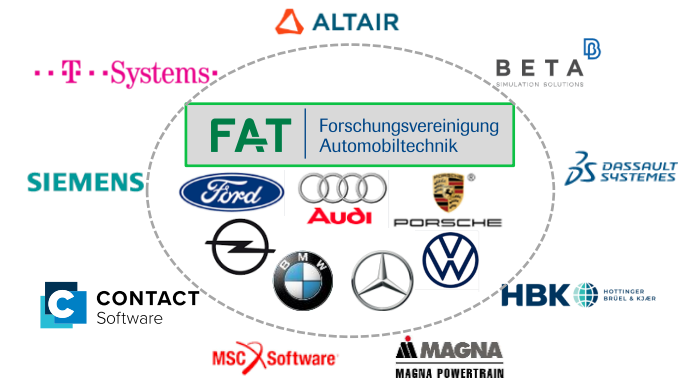
VDA | **FAT** | Forschungsvereinigung Automobiltechnik | **AK 25** | Fügetechnik | χMCF WG defines and maintains χMCF.

XML-based χMCF meets all “usual” requirements to a standard (**incl. long-time ability!**), plus:

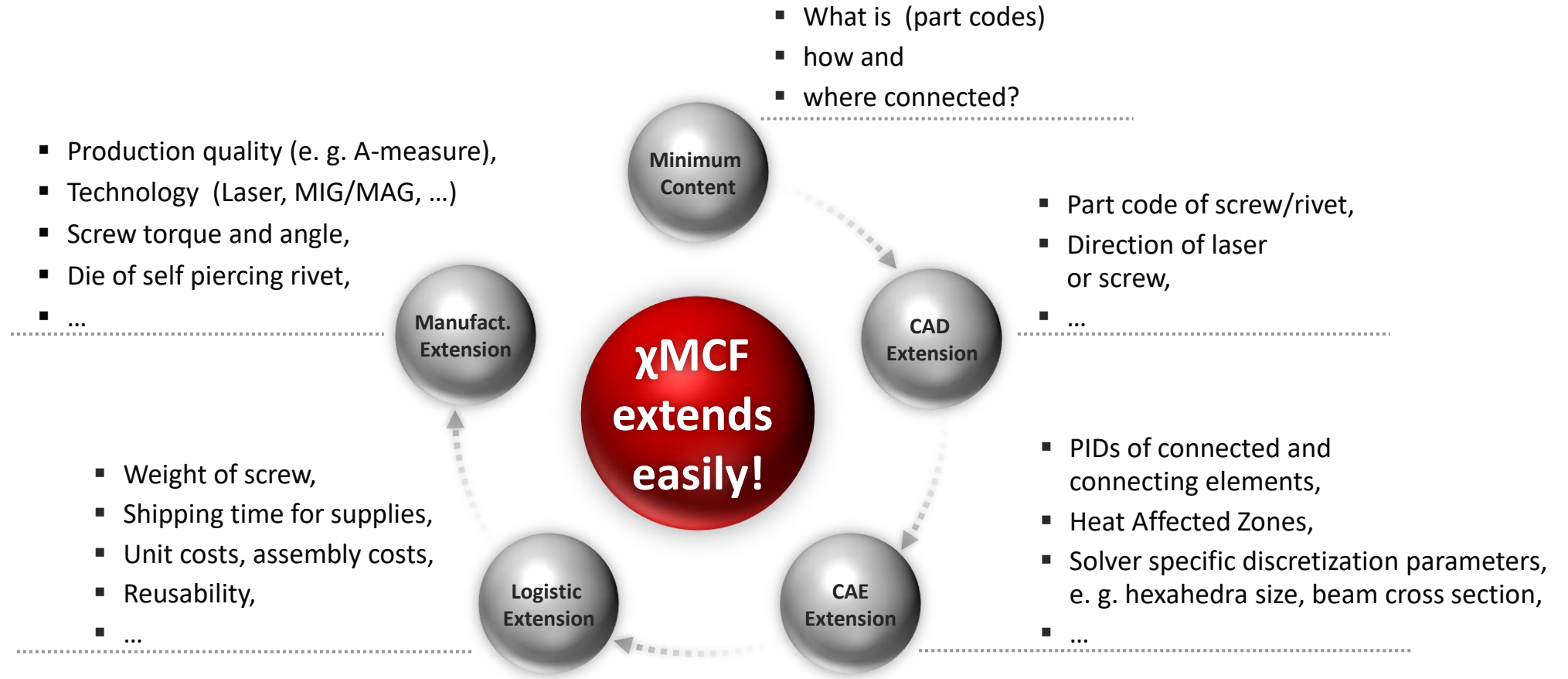
- All connection types & techniques can be represented.
 - Unambiguous, completely, exact, and to the current design maturity.
- All PLM processes are supported – CAD, CAE, CAT, Manufacturing Planning & Execution, including special sub processes, e. g.:
 - Durability simulation,
 - Robot programming,
 - Supplier integration, ...
- One χMCF-file contains either data of one *assembly*, one *car* or all *variants* of a series.
⇒ χMCF meets any kind of *OEM specific process* design.
- χMCF allows imbedding *custom data* specific to *OEM*, *process* and *tool*.

⇒ All existing proprietary formats can be replaced sustainably.

- VDA: German Association of the Automotive Industry
- FAT: Research Association for Automotive Technology; department of VDA
- AK 25: Working Group 25; focus on joining technologies



χMCF Accumulates Data along the Process Chain

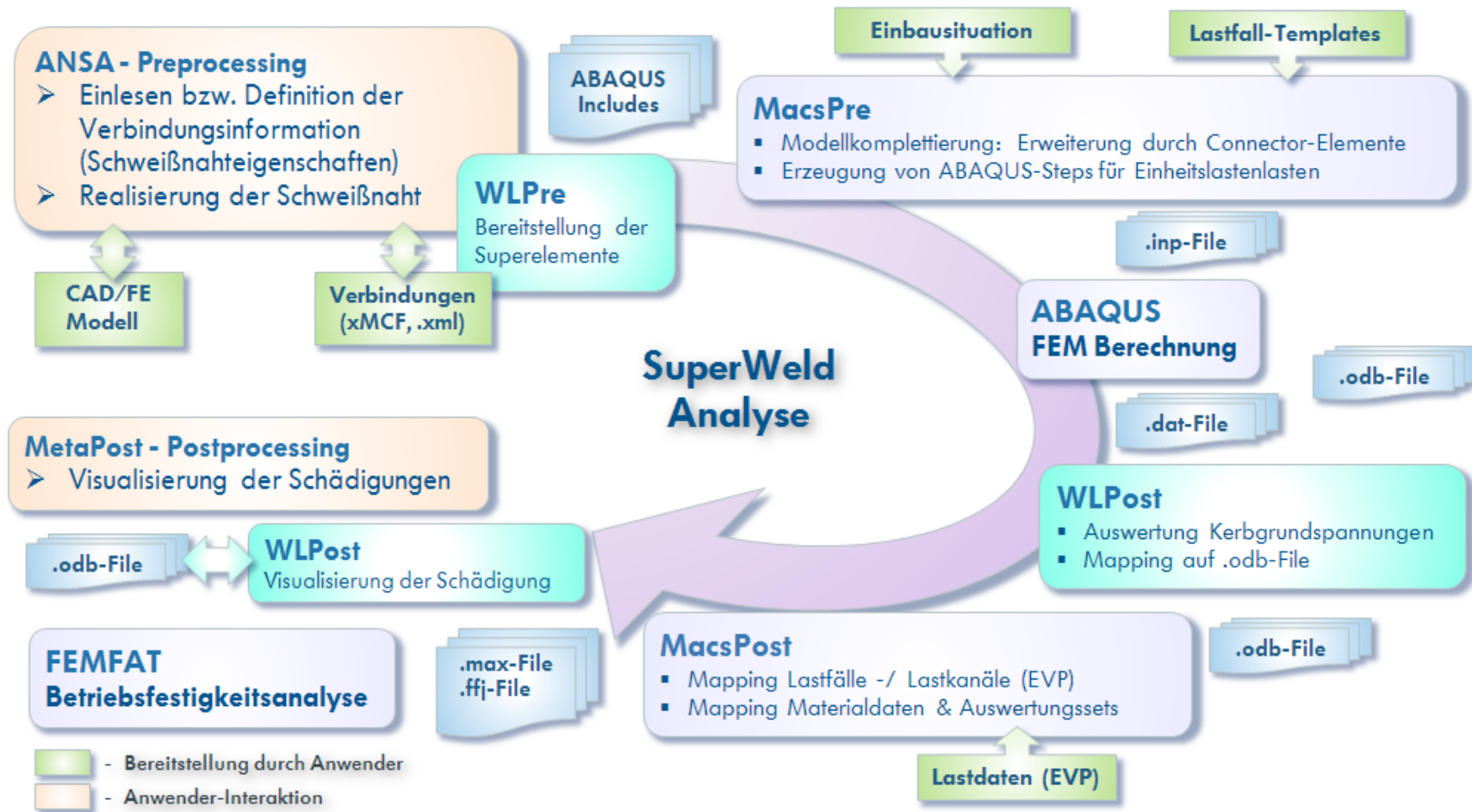


χ MCF Use Case CAE Integration



Use Case 0: CAE Integration

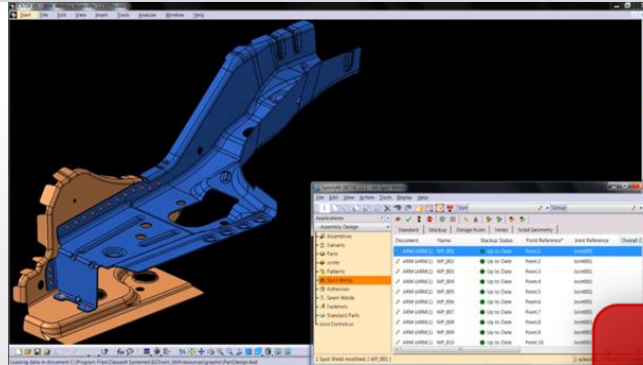
Variant of a fatigue process at Volkswagen, 2014



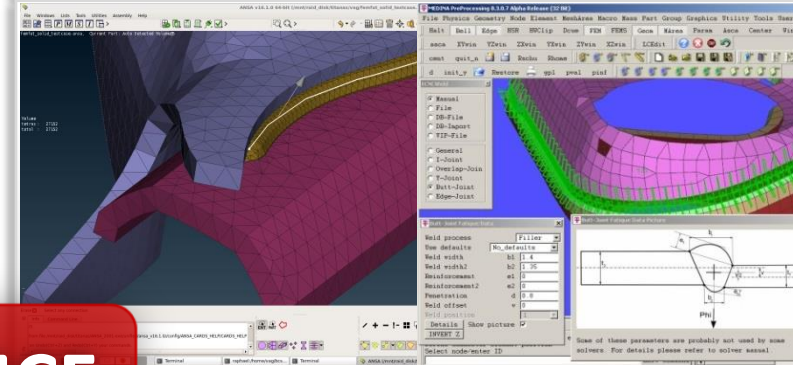
Many different tools.
Many different formats.

χ MCF-based Tool-Integration (as of 2017)

CAD e.g. SyncroFIT for CATIA & NX

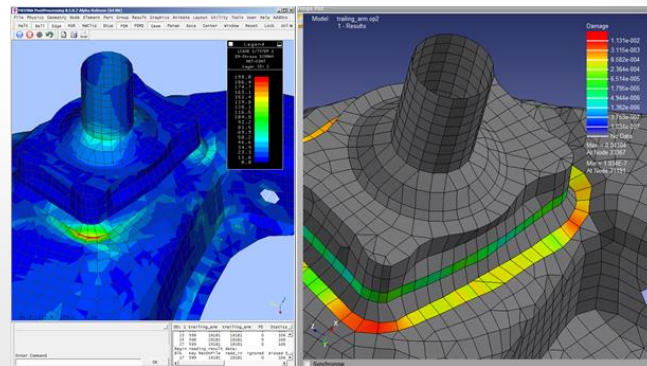


CAE Pre e.g. ANSA, MEDINA

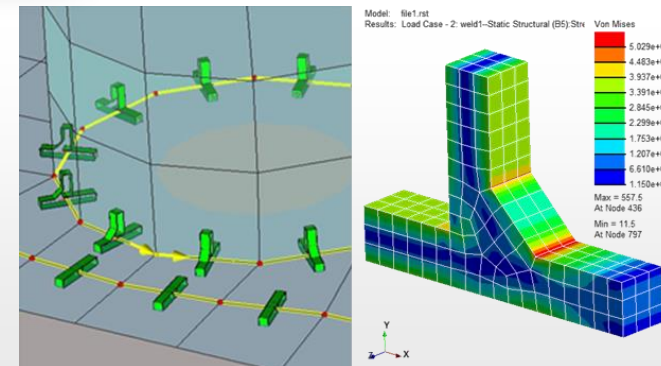


χ MCF

CAE Post e.g. MEDINA, nCode



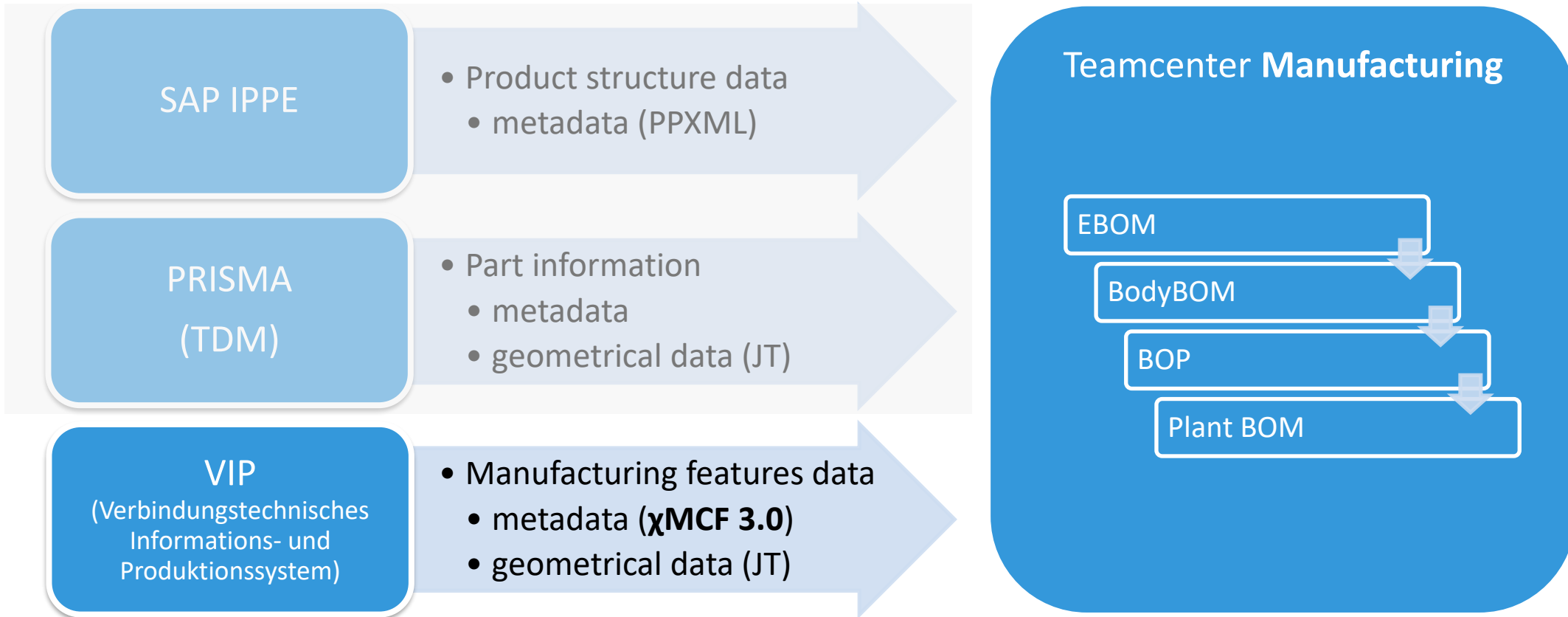
CAE Solve e.g. LMS, nCode, FEMFAT



Strategy & Example χ MCF Use Cases at **BMW**



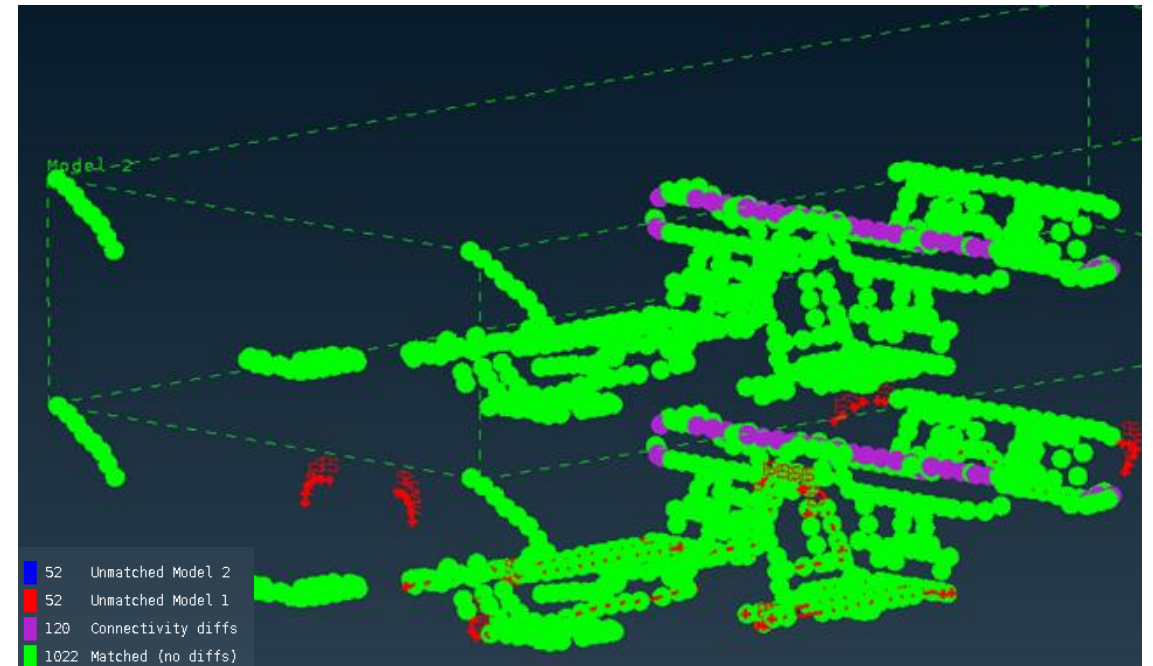
Use Case @ BMW (2018): χ MCF Data Exchange between PDM & **Production Planning**



BMW χ MCF Strategy

- **Production Planning**
(Teamcenter Manufacturing)
was addressed 2018.
- **CAE** was addressed 2019.
Data for CAE contains
technology parameters,
e.g. weld shape.

FEM preprocessor acted as a *verification tool*,
comparing VIP & χ MCF data
during prototype phase:



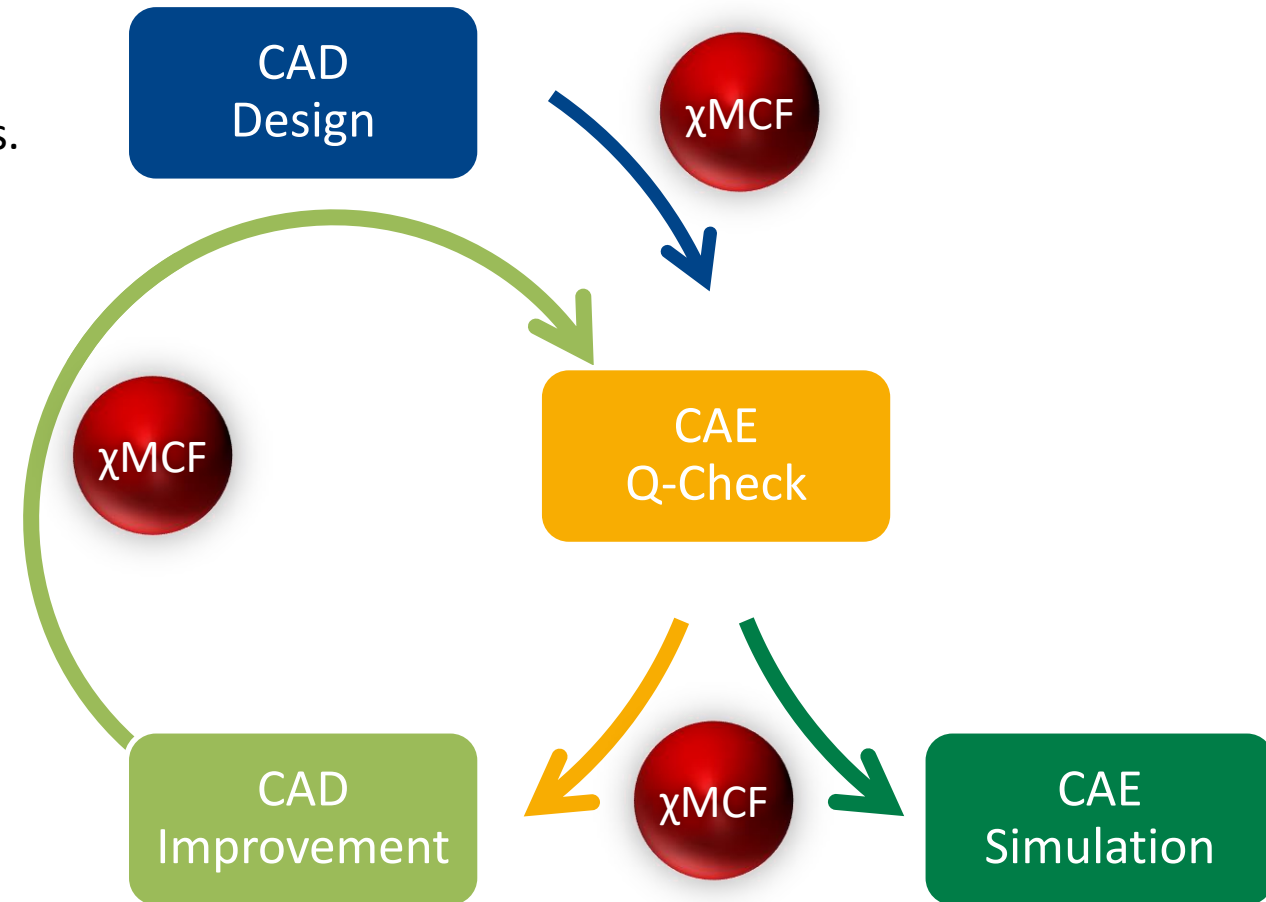
χ MCF Example Use Case at Volkswagen



Use Case @ at Volkswagen (2019):

Quality-Gate between CAD and CAE – Enabled by χ MCF

- Frequently, a complete digital vehicle is assembled *in CAE for the first time* in product development process.
- Using χ MCF, connection data can be provided to CAE in the most *automated and low-error fashion*.
- Powerful features of FEM a preprocessor allow for *automated, fast and reliable quality checks*.
- Custom scripts provide *custom error categorization*.
- Via χ MCF, categorized quality issues can be *sent back to design*.
- Categories allow CAD to fix the issues by a defined and *plannable process*.



Use Cases of prostep ivip's PDM-IF: **χMCF and STEP AP 242** federated

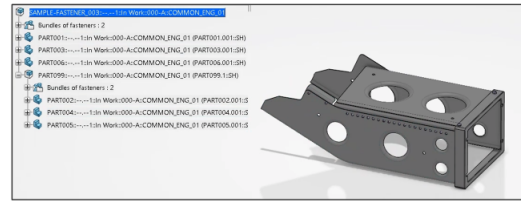
- Dassault Aviation
- BMW

Quoted from prostep ivip's PDM Interoperability Forum (PDM-IF)

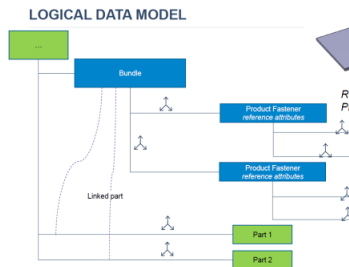
Use Case Status

Fasteners

- Full dataset ready for Dassault Aviation Use Case
 - Assembly Structure mapped from 3DExp->CATIA V5->AP242 XML Edition 3 + P21/JT
 - Enriched with Bundle definition, references to χMCF file and OccurrenceRelationships between LinkedParts and ProductFasteners (parametric definition of the holes)
 - χMCF file containing the structure and the reference/instance attributes of the Bundles and the ProductFasteners
 - Mapping of the attributes to standard attributes or customized attributes
 - Beta CAE could read this dataset into ANSA



Forum

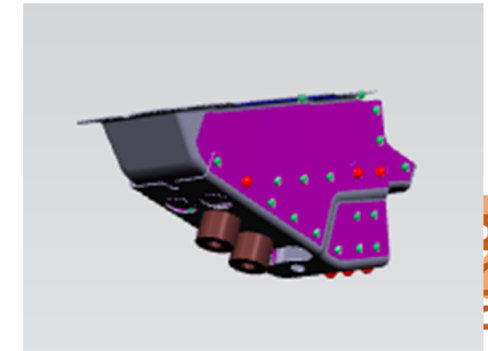


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Use Case Status

Fasteners

- Full dataset ready for BMW Use Case
- χMCF File provided by BMW
 - containing the structure and the reference/instance attributes of Carrier Sheets and Joining Elements
 - Many customized attributes seem to contain PDM data
- Assembly Structure (stpx) + Geometry (JT)
 - Enriched with Fasteners and reference to χMCF file
 - Beta CAE could read this dataset into ANSA
- Analysis of the used χmcf custom_attributes
 - Proposal where to move some of them to AP242 XML file and complete the dataset with combination of AP242 and χMCF



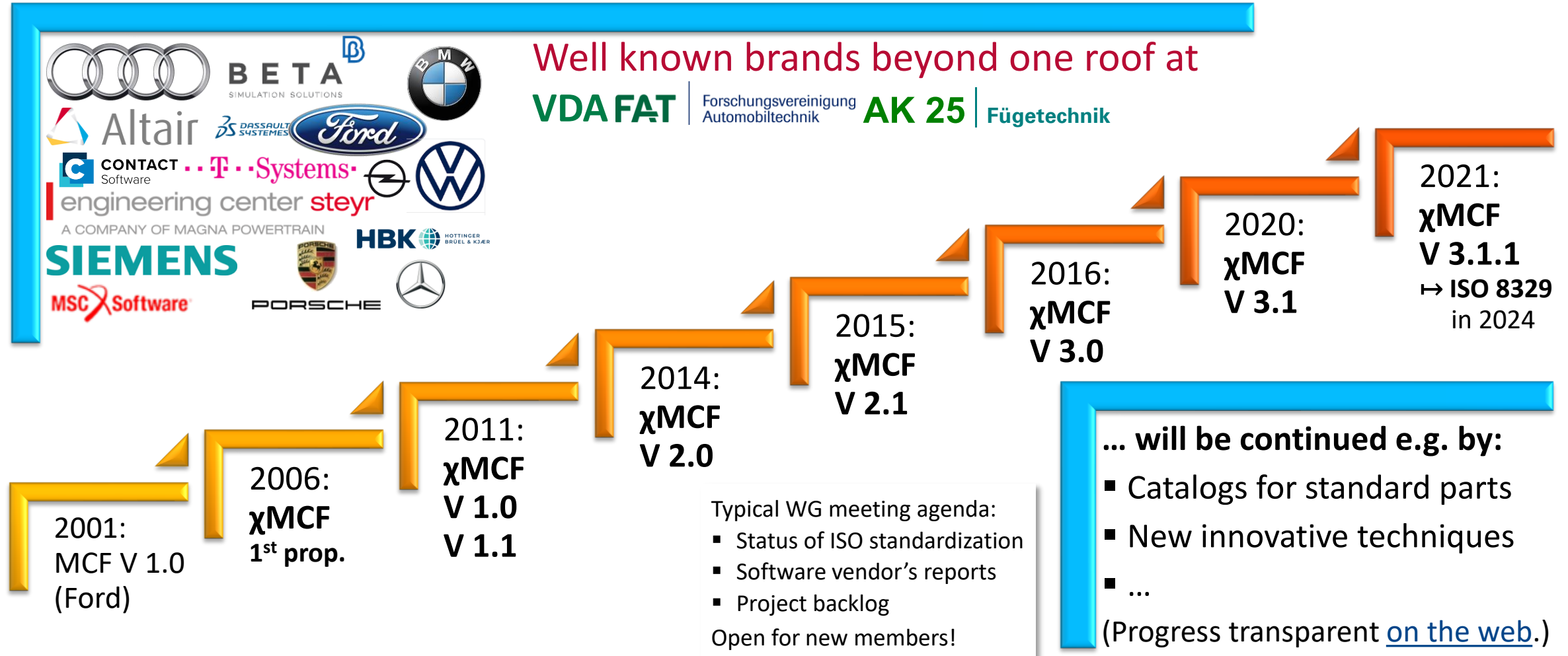
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χMCF – A Standard with History and Broad Support



Summary

- Connection processes are *rich and manifold* – so are the data.
 - χ MCF is *the* powerful and mature standard for piping connection data forward and backward through the product development process.
 - It is able to *bridge any gap* between process steps or tools.
 - As has been shown by applications at BMW & Volkswagen.
 - Many *important tools already support* χ MCF – more to come.
- ⇒ AK 25 provides support when *optimizing processes to benefit* from χ MCF.
- ⇒ More *demand placed at software vendors* will lead to even wider support of χ MCF.



Current Status of Standardization

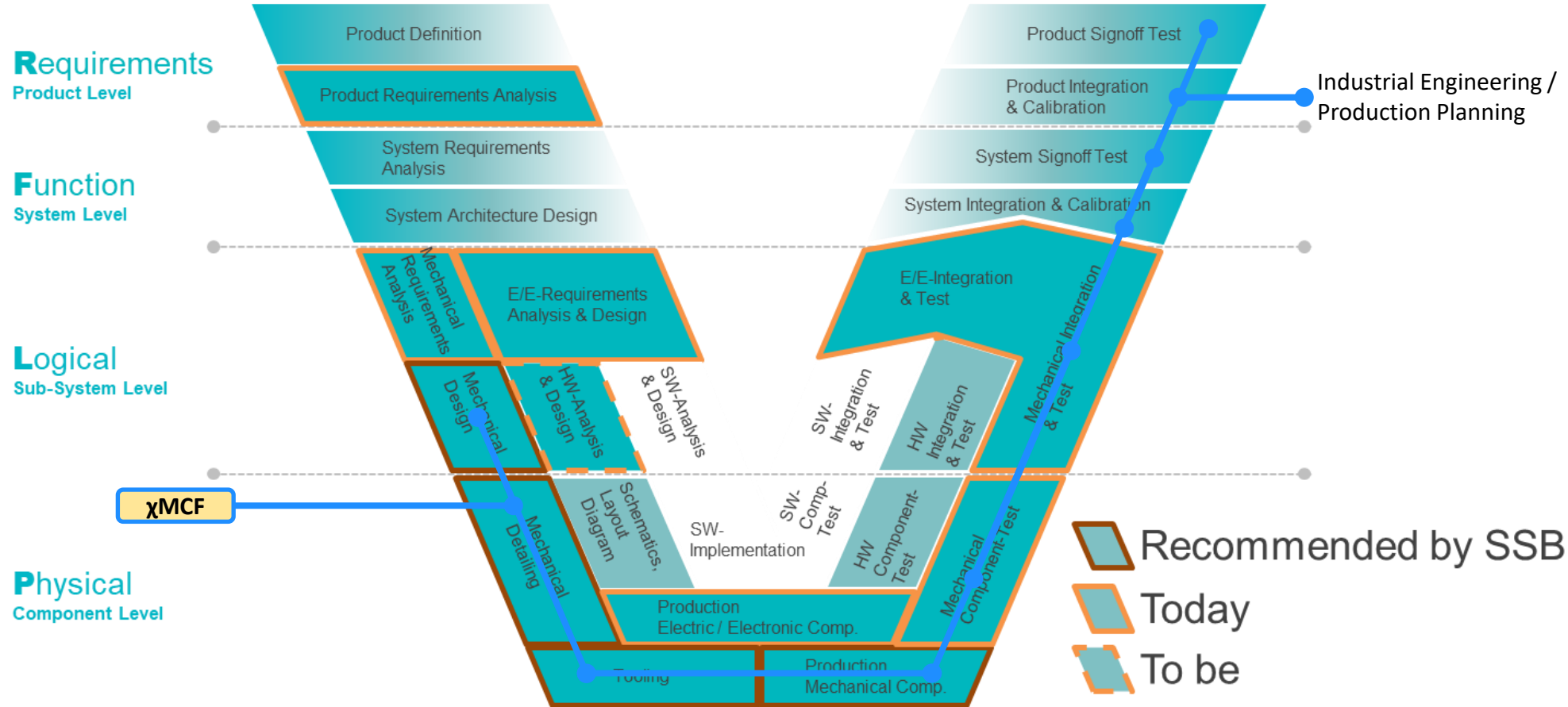
- ISO/PAS 8329 describes χ MCF 3.1.1.
- It was published 2024-08-06.
- VDA working group discusses further progress, e.g. just yesterday (2025-10-15).

	FINAL DRAFT Publicly Available Specification
Extended master connection file (χMCF) — Description of mechanical connections and joints in structural systems	ISO/DPAS 8329 ISO/TC 184/SC 4 Secretariat: ANSI Voting begins on: 2024-06-10 Voting terminates on: 2024-08-05 RECIPIENTS OF THIS DRAFT ARE INVITED TO SUBMIT, WITH THEIR COMMENTS, NOTIFICATION OF ANY RELEVANT PATENT RIGHTS OF WHICH THEY ARE AWARE AND TO PROVIDE SUPPORTING DOCUMENTATION. IN ADDITION TO THEIR EVALUATION AS BEING ACCEPTABLE FOR INDUSTRIAL, TECHNO- LOGICAL, COMMERCIAL AND USER PURPOSES, DRAFT INTERNATIONAL STANDARDS MAY ON OCCASION HAVE TO BE CONSIDERED IN THE LIGHT OF THEIR POTENTIAL TO BECOME STANDARDS TO WHICH REFERENCE MAY BE MADE IN NATIONAL REGULATIONS. Reference number ISO/DPAS 8329:2024(en) © ISO 2024



Strategic Position of χ MCF as a Standard

Carsten Franke's personal assessment ...



χMCF as a Standard – Canvas

Carsten Franke's personal assessment ...

Short Description:

- χMCF describes mechanical / structural connections.
- Any type
- Any technique
- Any PDP process – from component- via product- to 150% model-based

Goals:

- Easy to read, even for humans.
- No redundancy with prod. tree & CAD.
- Easily expandable for
 - specific user groups / task types
 - tools / algorithms
- Timely adaptation to new joining techniques and simulation methods.

Promoting Bodies:

- ☑ ISO/PAS 8329
- ☑ VDA FAT-Schriftenreihe 286

Maturity:

- CAD & FE tools read & write χMCF.
- Productive Use Cases at BMW.
- Aerospace interested.
- Supported by well-known software vendors.
- Six Releases since 2011.
- Federated use with STEP AP 242 ed. 5.

Scope:

- Design,
 - Issue- / Change Management,
 - Structure simulation & optimization,
 - Industrialization & production,
 - and service
- ... of structural connections.

Relevance for PDP:

- χMCF originated in fatigue simulation,
- addressed V&V-requirements from the very beginning,
- enables fast Q-Gate during data transfer CAD → CAE,
- Supports mechanical engineering from CAD to SOP and even service.

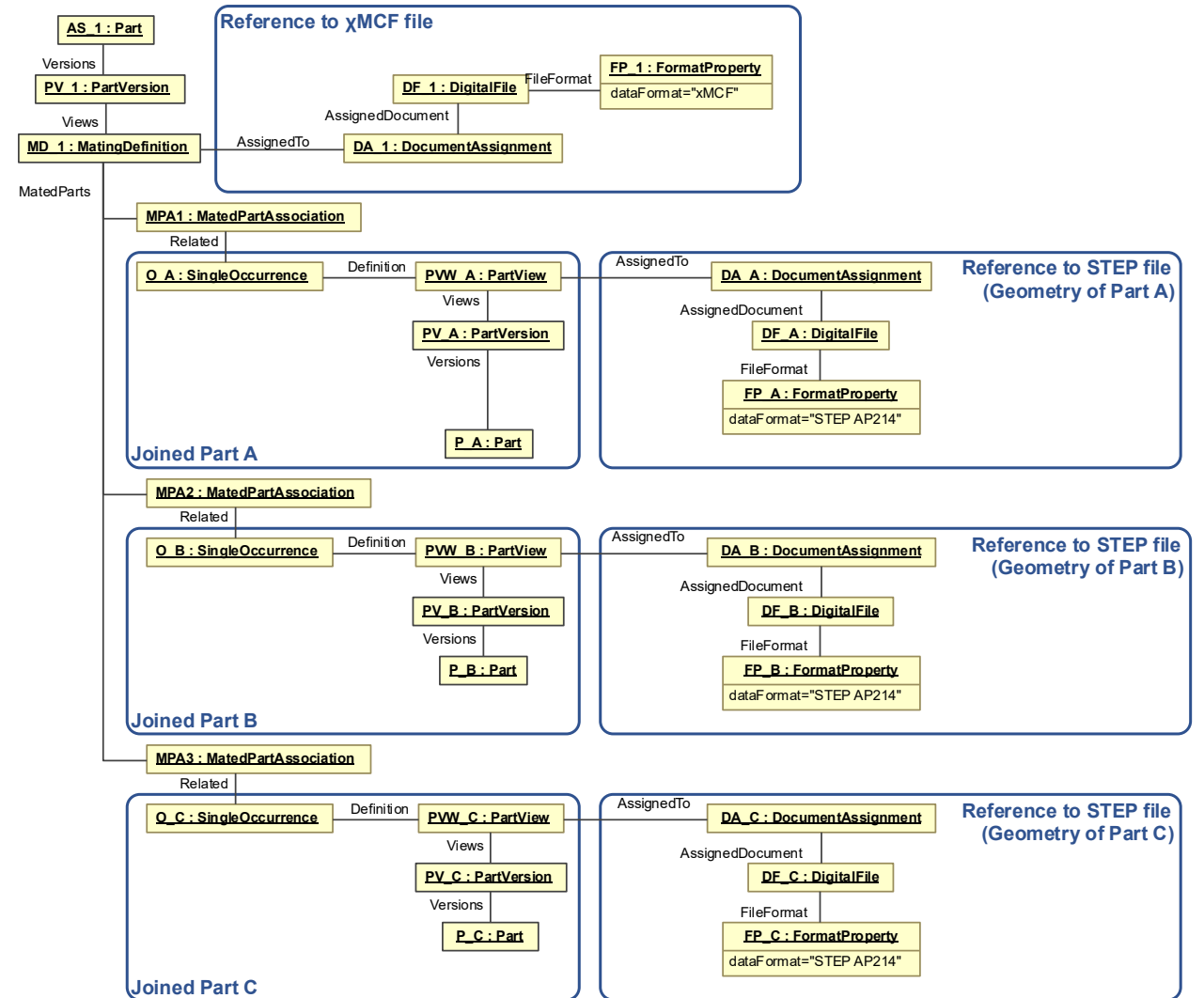
Federated use of STEP AP242 with χ MCF



Federating the Standards

Approach:

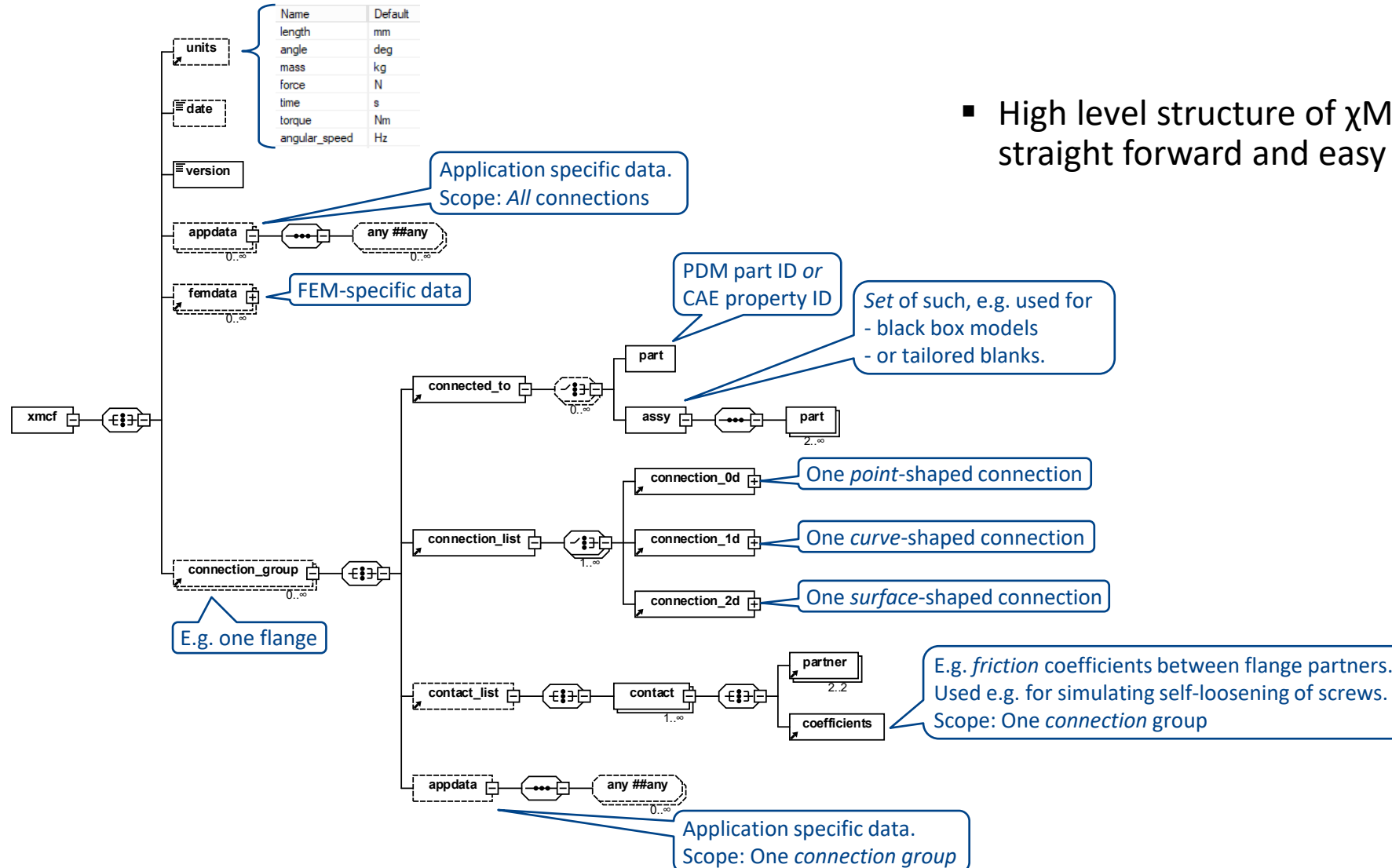
- Include *basic* mating information in AP 242 (existing capabilities)
- with linking to χ MCF, where χ MCF has a high level of details of the connection.



Technical Details of χ MCF

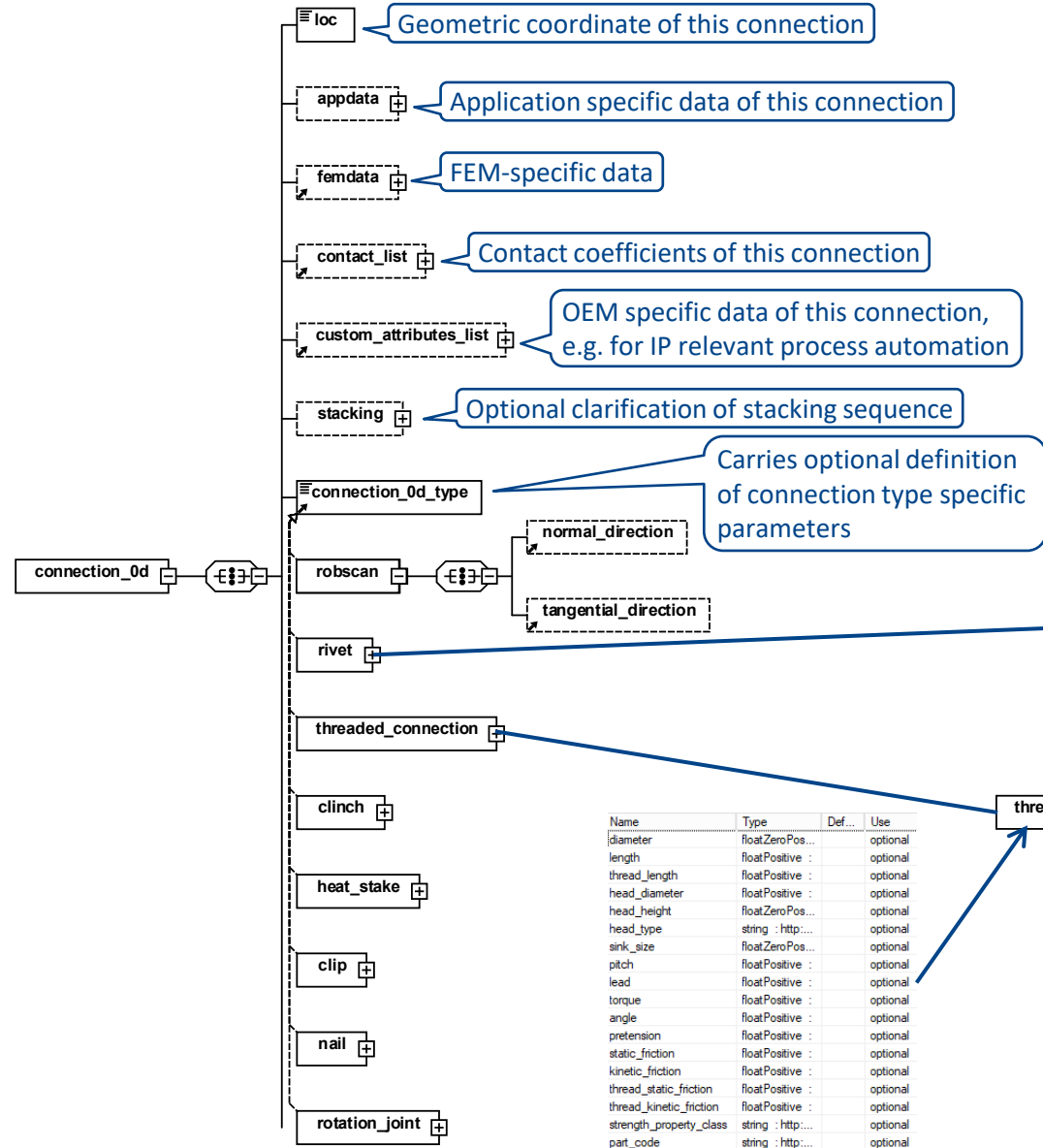


Structure of χ MCF 3.1.1 XSD – High Level



- High level structure of χ MCF is straight forward and easy to understand.

Structure of χ MCF 3.1.1 XSD – Point Connections

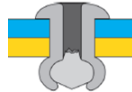


- For point connections, 8 types are supported. Each with its own, additional parameters.
- Some have sub-types and sub-elements, e.g. <threaded_connection/>.

Name	Type	Default	Use
hardness	floatPositive		optional
head_diameter	floatPositive		optional
head_height	floatZeroPositive		optional
head_type	string	http://ww...	optional
length	floatPositive		optional
part_code	string	http://ww...	optional
shaft_diameter	floatPositive		optional
sink_size	floatZeroPositive		optional
strength_property_class	string	http://ww...	optional

normal_direction

rivet_type



blind

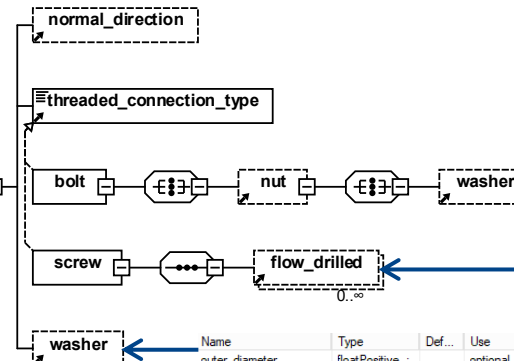
self_piercing

solid

swop

clinch_rivet_stud

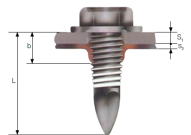
Name	Type	Default	Use
hole_depth	floatPositive		optional
hole_diameter	floatPositive		optional
max_grip	floatPositive		optional
min_grip	floatPositive		optional
shoulder_diameter	floatPositive		optional
shoulder_length	floatPositive		optional
tennon_diameter	floatPositive		optional
tennon_length	floatPositive		optional



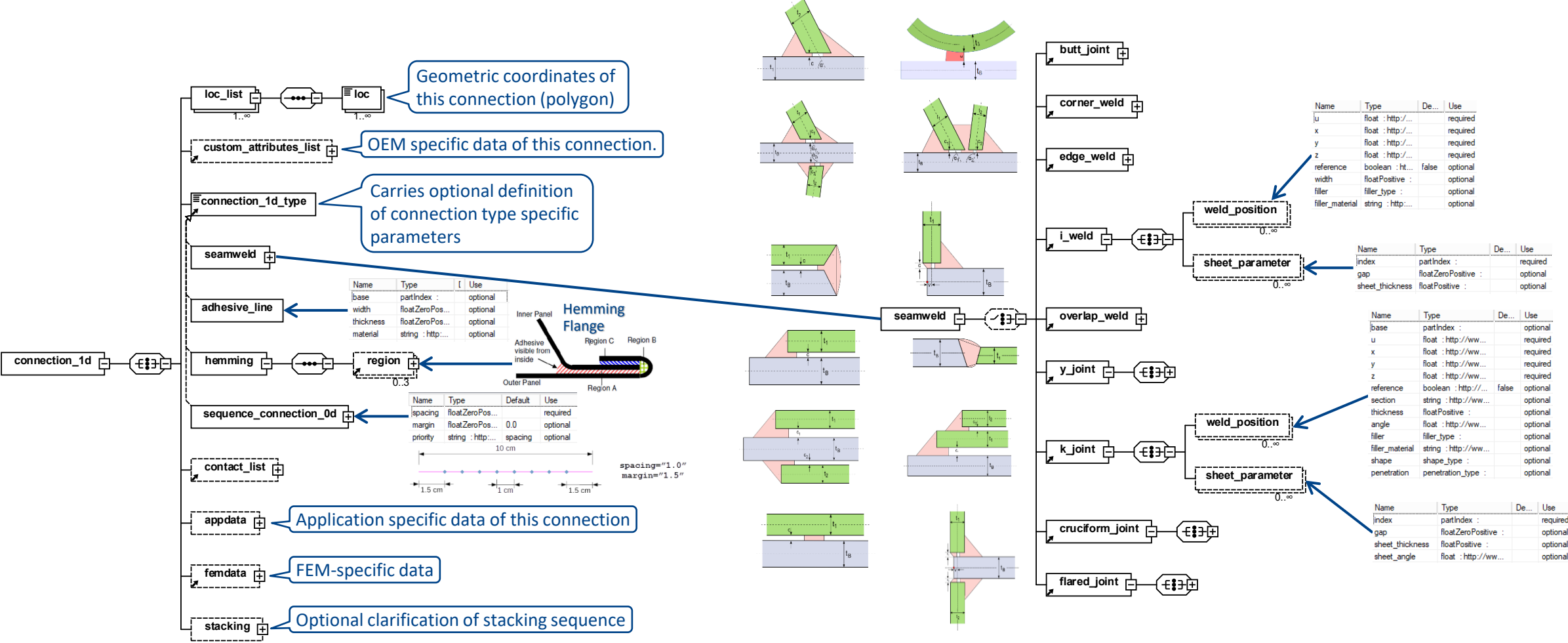
Name	Type	Def...	Use
diameter	floatZeroPos...		optional
length	floatPositive		optional
thread_length	floatPositive		optional
head_diameter	floatPositive		optional
head_height	floatZeroPos...		optional
head_type	string	http...	optional
sink_size	floatZeroPos...		optional
pitch	floatPositive		optional
lead	floatPositive		optional
torque	floatPositive		optional
angle	floatPositive		optional
pretension	floatPositive		optional
static_friction	floatPositive		optional
kinetic_friction	floatPositive		optional
thread_static_friction	floatPositive		optional
thread_kinetic_friction	floatPositive		optional
strength_property_class	string	http...	optional
part_code	string	http...	optional

Name	Type	Def...	Use
outer_diameter	floatPositive		optional
inner_diameter	floatPositive		optional
thickness	floatPositive		optional
attached	boolean	ht...	optional
static_friction	floatPositive		optional
kinetic_friction	floatPositive		optional
strength_property_class	string	http...	optional
part_code	string	http...	optional

Name	Type	f	Use
pre_machined_hole_diameter	floatZeroPos...		optional
pre_machined_hole_index	partIndex		optional
pilot_hole_diameter	floatZeroPos...		optional

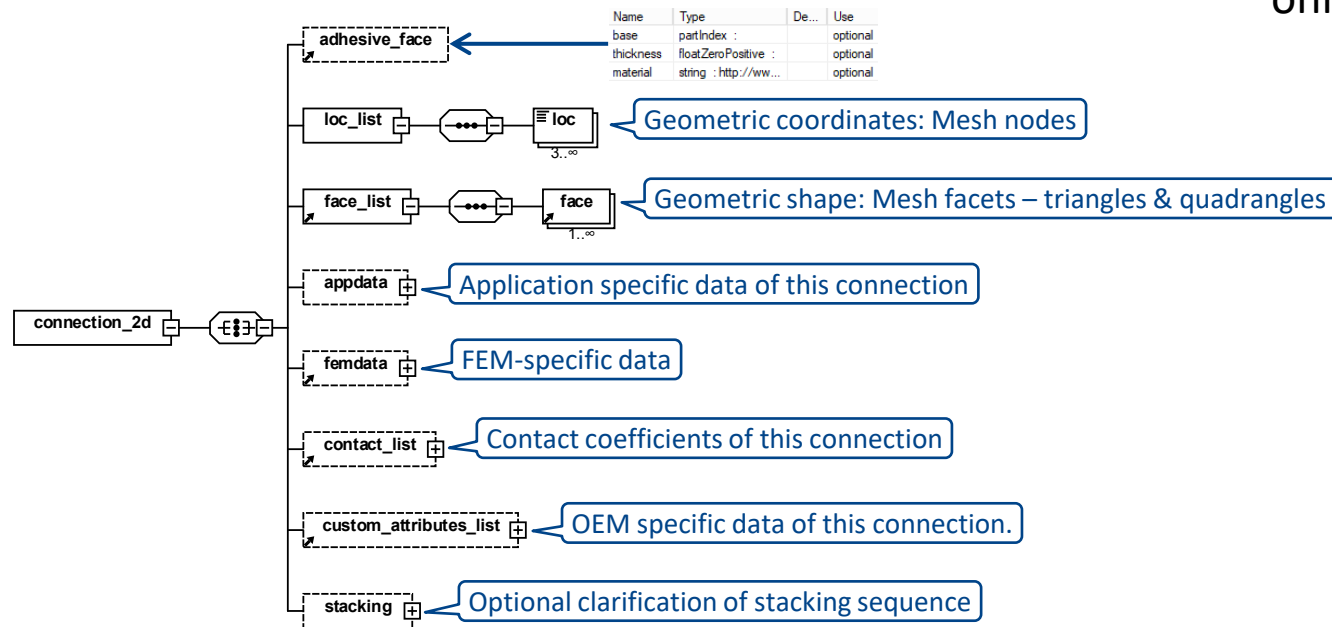


Structure of χ MCF 3.1.1 XSD – Curve Connections



Structure of χ MCF 3.1.1 XSD – Surface Connections

- For surface connections, only one type is defined: adhesive gluing.



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